11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

ANS:

11.28

a.

Demand equation from (11.11):

$$Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$$

Rewrite

$$\Rightarrow \alpha_2 P_i = Q_i - \alpha_1 - \alpha_3 PS_i - \alpha_4 DI_i - e_{di}$$

$$\Rightarrow P_i = -\frac{\alpha_1}{\alpha_2} + \underbrace{\overset{-}{\frac{1}{\alpha_2}}} Q_i \underbrace{- \overset{+}{\frac{\alpha_3}{\alpha_2}}} PS_i \underbrace{- \overset{+}{\frac{\alpha_4}{\alpha_2}}} DI_i - \frac{1}{\alpha_2} e_{di}$$

$$\begin{cases} \frac{1}{\alpha_2} < 0 \Rightarrow \alpha_2 < 0 \\ \alpha_3 > 0 \Rightarrow -\frac{\alpha_3}{\alpha_2} > 0 \\ \alpha_4 > 0 \Rightarrow -\frac{\alpha_4}{\alpha_2} > 0 \end{cases}$$
#

⇒ demand curve 中 P和Q為反向關係，故Q的係數預期為負

Supply equation from (11.12):

$$Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$$

Rewrite

$$\Rightarrow \beta_2 P_i = Q_i - \beta_1 - \beta_3 PF_i - e_{si}$$

$$\Rightarrow P_i = -\frac{\beta_1}{\beta_2} + \underbrace{\overset{+}{\frac{1}{\beta_2}}} Q_i \underbrace{- \overset{+}{\frac{\beta_3}{\beta_2}}} PF_i - \frac{1}{\beta_2} e_{si}$$

$$\begin{cases} \frac{1}{\beta_2} > 0 \Rightarrow \beta_2 > 0 \\ \beta_3 < 0 \Rightarrow -\frac{\beta_3}{\beta_2} > 0 \end{cases}$$
#

⇒ supply curve 中 P. Q 為正向關係. 故 Q的係數預期為正 #

b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

ANS:

```
> fit_demand_iv <- ivreg(p ~ q + ps + di | ps + di + pf, data = dat)
> summary(fit_demand_iv)

Call:
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-39.661  -6.781   2.410   8.320  20.251 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -11.428     13.592  -0.841  0.40810    
q             -2.671      1.175  -2.273  0.03154 *  
ps             3.461      1.116   3.103  0.00458 ** 
di            13.390      2.747   4.875 4.68e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567,	Adjusted R-squared: 0.5056 
Wald test: 17.37 on 3 and 26 DF,  p-value: 2.137e-06
```

```
> fit_supply_iv <- ivreg(p ~ q + pf | ps + di + pf, data = dat)
> summary(fit_supply_iv)

Call:
ivreg(formula = p ~ q + pf | ps + di + pf, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.7983 -2.3440 -0.6281  2.4350 11.1600 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -58.7982     5.8592  -10.04 1.32e-10 ***
q             2.9367     0.2158   13.61 1.32e-13 ***
pf            2.9585     0.1560   18.97  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486,	Adjusted R-squared: 0.9448 
Wald test: 232.7 on 2 and 27 DF,  p-value: < 2.2e-16
```

The first figure above shows the regression result of demand equation. One can see that the coefficients for quantity(q), price for substitutions(ps), and disposal income(di) are negative, positive, and positive respectively, which are in accordance with the signs from (a). Furthermore, since the p-value for these three variables are smaller than the significance level 5%, as a result, we can infer that the estimated coefficients are significantly different from zero.

The second figure above shows the regression result of supply equation. One can see that the coefficients for quantity(q), and price for factor of production(pf) are both positive, which are in accordance with the signs from (a). Furthermore, since the p-value for these two variables are smaller than the significance level 5%, as a result, we can infer that the estimated coefficients are significantly different from zero.

c. Estimate the price elasticity of demand "at the means" using the results from (b).

ANS:

```
> b_q_demand <- coef(fit_demand_iv)["q"]
> avg_p <- mean(dat$p)
> avg_q <- mean(dat$q)
> elasticity <- (1 / b_q_demand) * (avg_p / avg_q)
> cat("\n(c) 在平均值處的需求價格彈性:", elasticity, "\n")

(c) 在平均值處的需求價格彈性: -1.272464
```

One can see that the elasticity of demand is -1.272464, whose absolute value is greater than one means elastic. Therefore, when price increases one percent, the quantity demanded will decrease approximately 1.27 percent at the means.

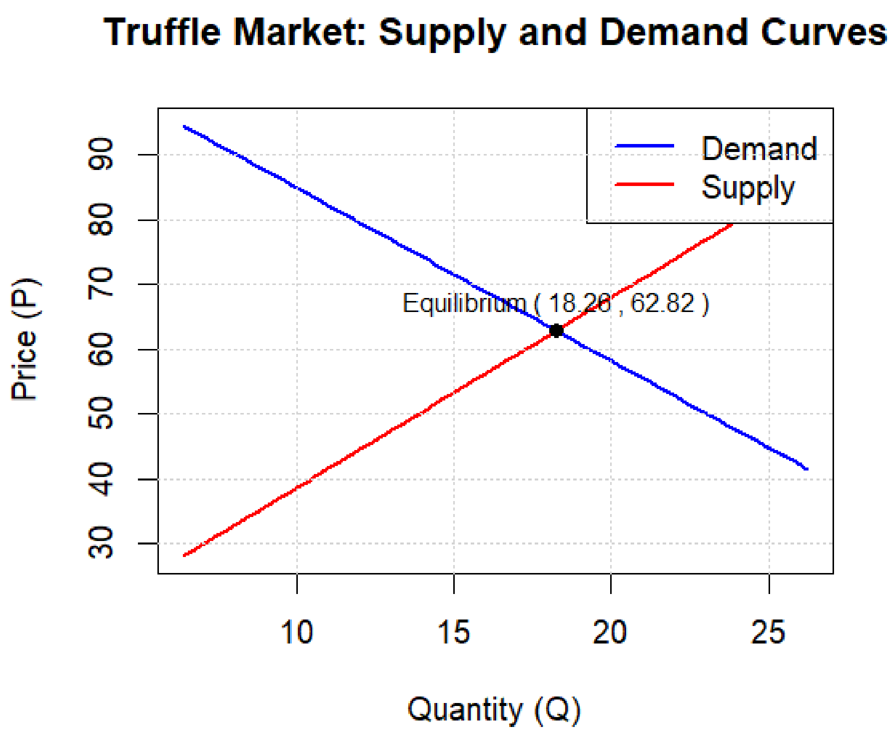
d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI, PS, and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

```
> di_star <- 3.5
> pf_star <- 23
> ps_star <- 22
> red_p <- lm(p ~ ps + di + pf, data = dat)
> red_q <- lm(q ~ ps + di + pf, data = dat)
> new_data <- data.frame(ps = ps_star, di = di_star, pf = pf_star)
> p_equil <- predict(red_p, new_data)
> q_equil <- predict(red_q, new_data)
>
> cat("\n(e) 預測均衡價格:", p_equil, "\n")

(e) 預測均衡價格: 62.81537
> cat("\n(e) 預測均衡數量:", q_equil, "\n")

(e) 預測均衡數量: 18.2604
```

```
> # 1. 提取 (b) 小題 2SLS 估計的係數(含所有的變數)
> c_d <- coef(fit_demand_iv)
> c_s <- coef(fit_supply_iv)
> # 2. 設定 (d) 小題給定的外生變數固定值
> DI_star <- 3.5
> PF_star <- 23
> PS_star <- 22
> # 3. 計算截距項 (把固定值帶入方程式)
> # 需求曲線: P = alpha1 + alpha_q*Q + alpha_ps*PS + alpha_di*DI
> # 整理成 P = Intercept + Slope*Q 的形式
> intercept_d <- c_d["(Intercept)"] + c_d["ps"] * PS_star + c_d["di"] * DI_star
> slope_d      <- c_d["q"]
> # 供給曲線: P = beta1 + beta_q*Q + beta_pf*PF
> intercept_s <- c_s["(Intercept)"] + c_s["pf"] * PF_star
> slope_s      <- c_s["q"]
> # 4. 定義 Q 的範圍 (根據資料範圍設定，例如 0 到 100)
> q_axis <- seq(min(dat$q), max(dat$q), length.out = 100)
> # 5. 計算對應的 P 值(把不同的q值代入需求與供給函數找點)
> p_demand <- intercept_d + slope_d * q_axis
> p_supply <- intercept_s + slope_s * q_axis
> # 6. 開始繪圖
> plot(q_axis, p_demand, type = "l", col = "blue", lwd = 2,
+      ylim = c(min(p_demand, p_supply), max(p_demand, p_supply)),
+      xlab = "Quantity (Q)", ylab = "Price (P)",
+      main = "Truffle Market: Supply and Demand Curves")
> lines(q_axis, p_supply, col = "red", lwd = 2)
> # 7. 標註均衡點 (e小題的結果)
> # 這裡可以使用剛才計算的 p_equil 和 q_equil
> points(q_equil, p_equil, col = "black", pch = 19)
> text(q_equil, p_equil, labels = paste("Equilibrium (", round(q_equil,2), ",", round(p_equil,2), ")"),
+      pos = 3, cex = 0.8)
> # 8. 加入圖例
> legend("topright", legend = c("Demand", "Supply"),
+        col = c("blue", "red"), lwd = 2)
> grid() # 加入格線方便觀察
```



e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

ANS:

```
> # 1. 提取 2SLS 的結構係數 (從 b 小題的模型中)
> c_d <- coef(fit_demand_iv)
> c_s <- coef(fit_supply_iv)
> # 2. 設定 (d) 小題給定的外生變項數值
> PS_star <- 22
> DI_star <- 3.5
> PF_star <- 23
> # 3. 計算兩條曲線各自的「常數項」(Intercept + 其他固定外生項)
> # 需求曲線常數項 (A): alpha1 + alpha_ps*PS + alpha_di*DI
> A <- c_d["(Intercept)"] + c_d["ps"] * PS_star + c_d["di"] * DI_star
> # 供給曲線常數項 (B): beta1 + beta_pf*PF
> B <- c_s["(Intercept)"] + c_s["pf"] * PF_star
> # 4. 提取 Q 的斜率
> slope_d <- c_d["q"] # 這是 alpha_Q
> slope_s <- c_s["q"] # 這是 beta_Q
> # 5. 解聯立方程 (A + slope_d * Q = B + slope_s * Q)
> # 移項整理: Q * (slope_d - slope_s) = B - A
> # 因此 Q = (B - A) / (slope_d - slope_s)
> Q_star <- as.numeric((B - A) / (slope_d - slope_s))
> # 6. 將 Q_star 帶回任一式求 P_star (這裡帶入需求式)
> P_star <- as.numeric(A + slope_d * Q_star)
> # 7. 輸出結果
> cat("使用 2SLS 結構係數解出的均衡點：\n")
使用 2SLS 結構係數解出的均衡點：
> cat("均衡數量 Q* =", Q_star, "\n")
均衡數量 Q* = 18.25021
> cat("均衡價格 P* =", P_star, "\n")
均衡價格 P* = 62.84257
```

The equilibrium price is 62.82 and the equilibrium quantity is 18.26(using reduced-form equations), while the equilibrium price is 62.84 and the equilibrium quantity is 18.25(using 2SLS structural equations to calculate the equilibrium solution). As you can see, they agree perfectly. This perfectly demonstrates the mathematical consistency of the simultaneous equations model: the intersection of the structural equations yields the same equilibrium as the reduced-form predictions.

f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

ANS:

```
> # --- (f) 使用 OLS 估計並比較 ---
> fit_demand_ols <- lm(p ~ q + ps + di, data = dat)
> fit_supply_ols <- lm(p ~ q + pf, data = dat)
> summary(fit_demand_ols)

Call:
lm(formula = p ~ q + ps + di, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-25.0753  -2.7742  -0.4097   4.7079  17.4979 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -13.6195     9.0872  -1.499   0.1460    
q             0.1512     0.4988   0.303   0.7642    
ps            1.3607     0.5940   2.291   0.0303 *  
di           12.3582     1.8254   6.770 3.48e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared:  0.8013,	Adjusted R-squared:  0.7784 
F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09


> summary(fit_supply_ols)

Call:
lm(formula = p ~ q + pf, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.4721 -3.3287  0.1861  2.0785 10.7513 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
q             2.6613     0.1712   15.54 5.42e-15 ***
pf            2.9217     0.1482   19.71  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared:  0.9531,	Adjusted R-squared:  0.9496 
F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16

> 
> cat("\n(f) OLS 與 2SLS 比較 (需求):\n")

(f) OLS 與 2SLS 比較 (需求):
> print(cbind(OLS = coef(fit_demand_ols), IV = coef(fit_demand_iv)))
                   OLS         IV
(Intercept) -13.6194989 -11.428407
q             0.1512043  -2.670519
ps            1.3607045   3.461081
di           12.3582353  13.389921
```

```
(f) OLS 與 2SLS 比較 (供給):
> print(cbind(OLS = coef(fit_supply_ols), IV = coef(fit_supply_iv)))
                   OLS         IV
(Intercept) -52.876298 -58.798223
q             2.661281   2.936711
pf            2.921660   2.958486
```

The second figure shows that when using OLS to estimate the demand equation, the sign of quantity will be completely reversed, yet ps and di are still positive. What's more, from the first figure, one can see that the p-value for q is greater than the significance level 5%, while the p-value for ps and di are smaller than 5%. Therefore, the estimated coefficient for q is not significantly different from zero, and the estimated coefficient for ps and di are significantly different from zero.

The third figure shows that when using OLS to estimate the supply equation, the sign of quantity and pf are still positive(just same as 2SLS). Besides, the p-value for q and pf are much smaller than significance level 5%, which means the estimated coefficient for q and pf are significantly different from zero.

In conclusion, we can see that when using OLS to estimate the demand equation, it will produce positive demand curve, which violates the law of demand. This is just because of simultaneous equations' characteristic. Using OLS method will lead to biased and inconsistent estimation result. As a result, we should use 2SLS to get the right estimate.